TESLA

Back To The Future

INFOTAINMENT SYSTEM



ABOUT

To build an in-vehicle infotainment system for an Electric SUV of 2025 for Indian Millenial Segment mostly businessmen and executive level professionals. The car runs on Level 3 autonomy. How might this system evolve in Indian scenario? How might our co-pilot interact and engage/dis-engage with these new affordances? How might the system adapt to changing driving conditions?

The new interaction solves some key problems that we need to address as the technology is growing, unapologetically leaving behind human understanding to use it.

MAIN CONCERNS WITH CURRENT UI TREND







VOLVO XC40

Disconnect between features

Visual overload with display of so many features in a screen

Information hierarchy is missing

MERCEDES BENZ GLC

Heavy graphics in times when there is a ui revolution of minimalism in web and mobile apps (for ex, reflection and shadows)

Use of brand specific technological terms makes understanding tough

SKODA KODIAQ

Icons may not be understood individually as much as with descriptions

Icon / data visualisation heavy screens leads to vagueness



Unreliable

Luxury = Latest Technologies = More infomartion
User unable to focus on the road while engaging with info heavy UI



Time Consuming

Disconnected feature placement in the UI. Requires multi clicks to complete a task



Inconvenient

Current system persuades manual inputs from hand held devices



Apathetic

Not programmed to learn from user behaviour to make everyday tasks simpler

BEHAVIOUR OF TARGET AUDIENCE

Milennials are fast entering their prime spending years. Growing up in the age of rapid change- they have set expectations & priorities that are radically different from the older generation.
They want fun & convenience. Almost everything at touch of a button or gesture.

UNDERSTANDING USERS

Neil Desai



Age // 35 years
Occupation // Senior Business Head at an MNC
Family // Wife, 4 year old kid
Lives // A Duplex Apartment in a premium builder society in Delhi

"Life should be fun & comfortable
Work Harder & make everyday worth it."

Lifestyle

Loves his job
Loves to spend equal time with family mostly doing outdoor activities
Fitness Enthusiat
Collects expensive alcohol
Host Parties at his place
Likes to spend on branded wear
Eats at high end restaurants

Devices Owned

Apple TV
MacBook Pro
Herman Home Theatre
Google Home
Go Pro Camera
Latest iPhone

Aspirations

To be on the top of the latest technology.
To live in a Penthouse.
To reach the highest in employee pyramid at his workplace.
To give luxurious life to his family.

Personality

Success Driven
Socially Outgoing
Brand and Quality Conscious
Perfectionist
Eager to improve

MIND
Extrovert — Introvert

ENERGY
Observant — Intuitive

NATURE
Thinking — Feeling

TACTIC
Judging — Prospecting

SCENARIO 1

Weekdays

- Neil and his wife both work at corporate jobs.
- During the weekdays, he drives his car to first drop his wife at her office & then to his office.
- Mornings are rush time so they mostly take their breakfast in the car.
- They hate the toll road from delhi - gurgaon as its time consuming to get into queue to pay tolls everyday.
- Neil uses google maps to navigate every morning to dodge the traffic via alt routes.
- After dropping his wife & upon reaching his office, he gives his car to valet to save some time.
- At evening, while leaving from his office he intimates his wife about his ETA to pickup.
- He drives to his wife's workplace.
- He welcomes the wife, she takes over the driving seat.
- She adjusts the car environment according to her preferences like mirrors, seats, steering wheel, airflow.
- They drive home. Upon parking at home spot, they put the car on charge.

SCENARIO 2

Weekends

- Neil and his family are driving to Neil's parent's house in Chandigarh from Delhi on a weekend. (They go once a month over weekends)
- They are seated in front, kid in the rear seat. They insure to tie him in safety belt & put child lock on.
- The car is low on battery, they add a pitstop in the navigation to charging station.
- They play the 'the road trip' playlist and hit the road.
- The street has low light condition, to prevent collision, Neil struggles to stay alert for pedestrians crossing unannouncingly, bumpers and passing cars.
- They stop at food court on the highway and put the car on charge there.
- After dinner they are back on the road, choose easiest road for smooth drive, and turns ambient light on for the kid to sleep peacefully while they reach the destination late night.

UX GOALS

Primary

- Quick navigation
- Save time spent in traffic jams or toll lanes
- To keep a track of battery consumption and get choices on recharging on the way
- Play music playlist according to mood without much of a hassle

Secondary

- Control the environment of car at the touch of a button
- To be able to enjoy the ride withour any accidents (as is on indian roads)
- Easily share whereabouts with family and friends

FRUSTRATIONS

1. The fear of battery running out before reaching charging station
2. Switching between apps (For example : navigation and music player) when eyes are on road
3. Information overload in infotainment panel
4. Hates informing his whereabouts frequently
5. Hates stopping in queue for tolls

WHAT ARE PEOPLE SAYING?

"When driving on Indian roads, be aware that anyone can take over you from anywhere. The pedestrians cross the highways without announcing!"

"Lighting is so poor on the streets during the night, I prefer not to take the dark lanes"

"Buy a car only if it comes with a parking spot! There's no space anywhere for parking these days.."

"Paying for the annual car maintenance and motor insurance is more heavy on the pocket than the car itself!"

UNIQUE DEMAND OF INDIAN SCENARIO

1. A Road Assist tool to alert for obstacls like pedestrians, pits, speedy lane changers
2. Proper lane driving system in place to curb traffic jam
3. Since there are multiple routes to reach a destination, a system so intelligent to be able to guide and be trusted
4. On roads there is no or poor street lighting, a car should be self reliable
5. Difficulties in finding parking spots requires a smart way to help parking
6. Transparency and info on expensive car maintenance packages
7. Awareness of new vehicle features for easy adoption
8. Pollution free roads
9. Anti-theft system

USE CASES

1. ROAD ASSIST TOOL

- Scan road using active radar system
- Show 3D of non-connected objects
- Understand what is possible/active

2. LANE DRIVING SYSTEM

- Scan road using active radar system
- Visual Alert if not in lane
- Auto push gently towards other side to align in lane

3. TRAFFIC JAM

- Suggest Alt route
- Take input Yes/No
- Rerout/ Stay on the same route

4. POOR STREET LIGHTING

- Turn on dynamic headlight for better visibility
- Stay alert for obstacles
- Using AR, interpret visually what system is detecting
- Alert driver to stop/slow down

5. DIFFICULTY FINDING PARKING SPOT

- If car and parking zone is digitally connected, keep a track of parking occupancy virtually
- System connects with other cars arriving at the same time
- System finds if parking to be available upon your arrival

6. EXPENSIVE CAR MAINTENANCE

- Car records driver behaviour upon usage and handling
- Car records Vehicle health
- Car dealer and Insurance company gets a regular record of your this data
- They offer customised insurance and maintenance plans according to individual usage.

TASK FLOW

Open car & infotainment directly connects to Neil's Mobile

Tap on "Navigate to Office" button and starts driving to daily destination

Home (Delhi) — Via Wife's Office — Office (Gurgaon)

Voice command to play music - Morning playlist 1 (Phone Media)

Set surround sound to front seats

Cross toll booth, pay via car

Pick incoming call while navigating to office- Car switches to semi autonomous driving mode while on call

Drop wife & reach own office. Find parking & park via park assist/valet mode

Evening: Enter car & get weather & traffic updates

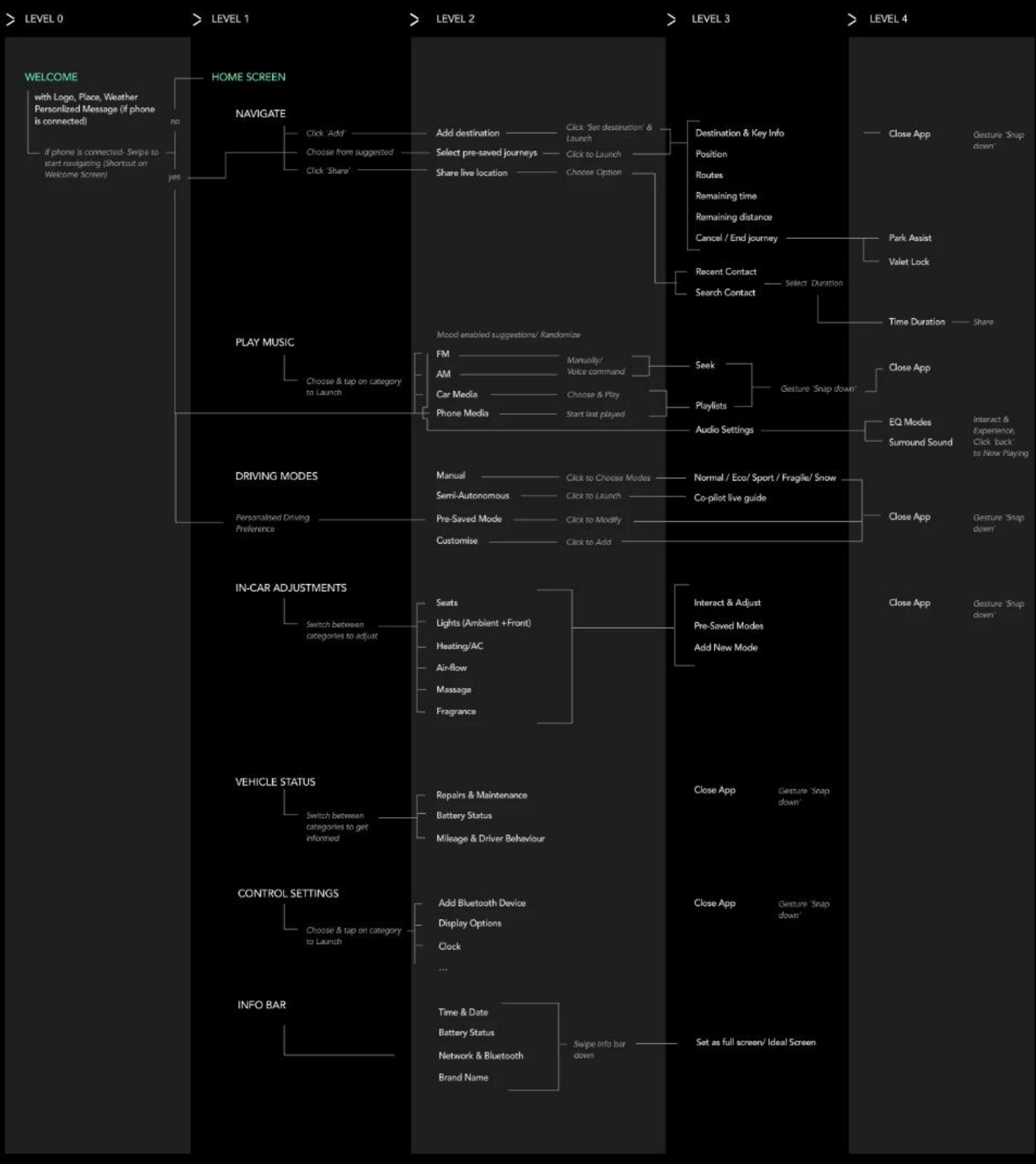
Notify wife for pick up and share live location

Pick up wife, switches position, wife takes driver seat

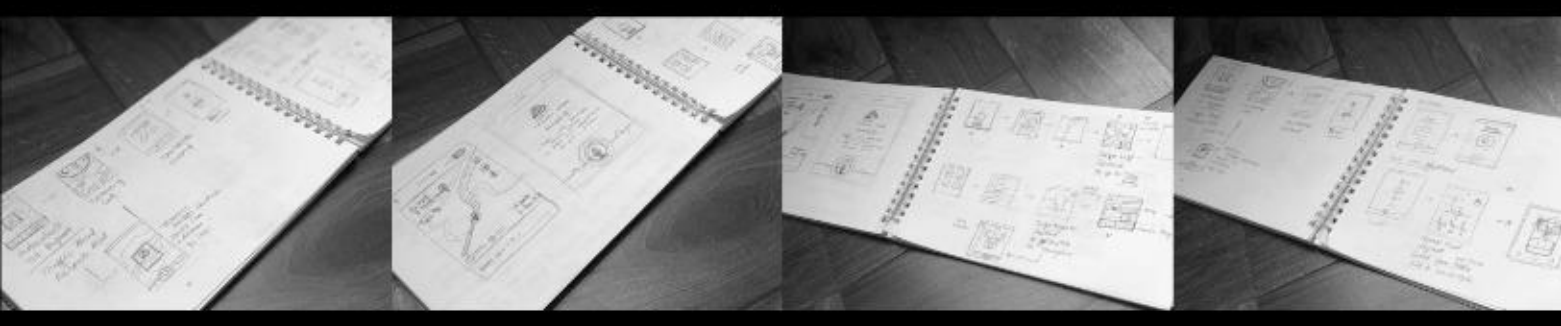
Car adjusts itself accordingly - mirrors and seats. Drive back home

INFORMATION ARCHITECTURE

Using logical flow a structure is created that draws hierarchies in 5 levels

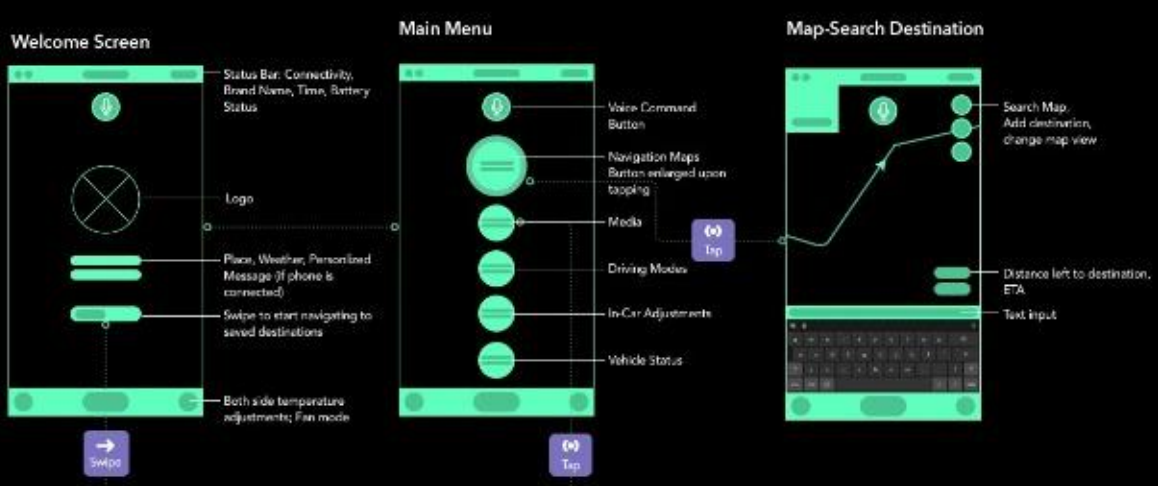


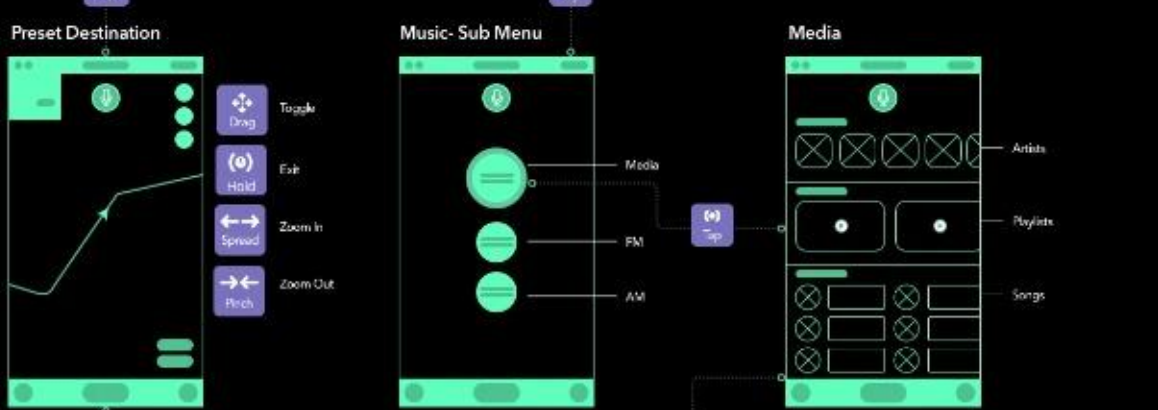
PAPER SKETCHES



SCREEN FLOW

Rough paper sketches of the the task flow are converted into digital low-fi screens













MAIN FEATURES OF SOLUTION

PROGRESSIVE DISCLOSURE

Information to be revealed one step at a time so that user gets mental space and time to digest the information

LESS IS MORE

To give more information and output in least required words and visuals

ADAPT OVER EXPECT

System to smartly adapt to Drivers inputs. For ex. Personalised experience upon phone connectivity

RELIABILITY

Expose the intelligence of system through what it sees and what it plans to do through better communication

EXPECTED USE OF TECHNOLOGY

GESTURE RECOGNITION

Gesture recognition reduces drivers distraction & enhances safety. Controlling In-Car Infotainment Systems Approaching your hand a few inches towards the infotainment system will activate its screen and present you with a bunch of relevant icons

AUGMENTED REALITY

To streamline relevant information, such as speed, turn-by-turn directions, or the track information of a currently-playing song, into the user's field of view

HAPTIC FEEDBACK

Haptic touchscreen feedback has improved user experience, driving performance and increased confidence in driver visual behaviour

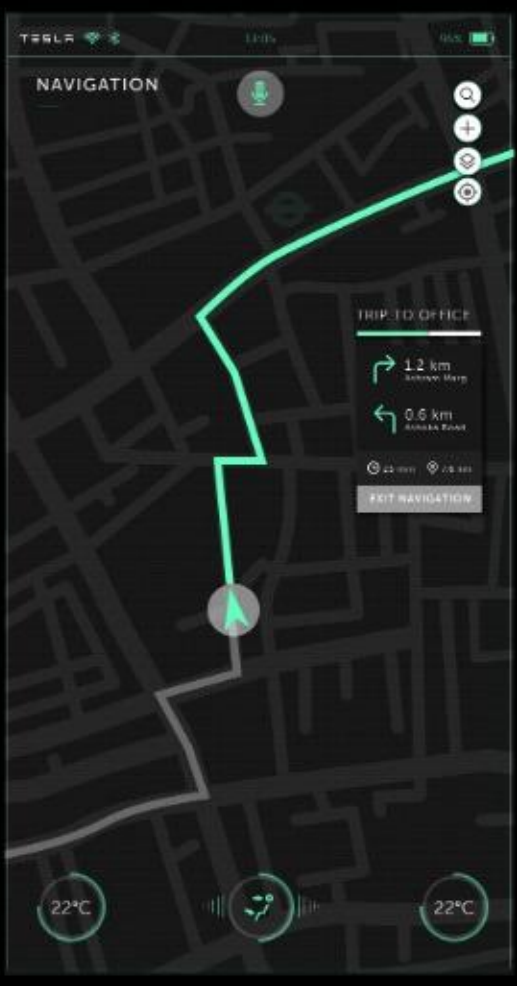
WIREFRAMES

Visual Representation of collective ideas

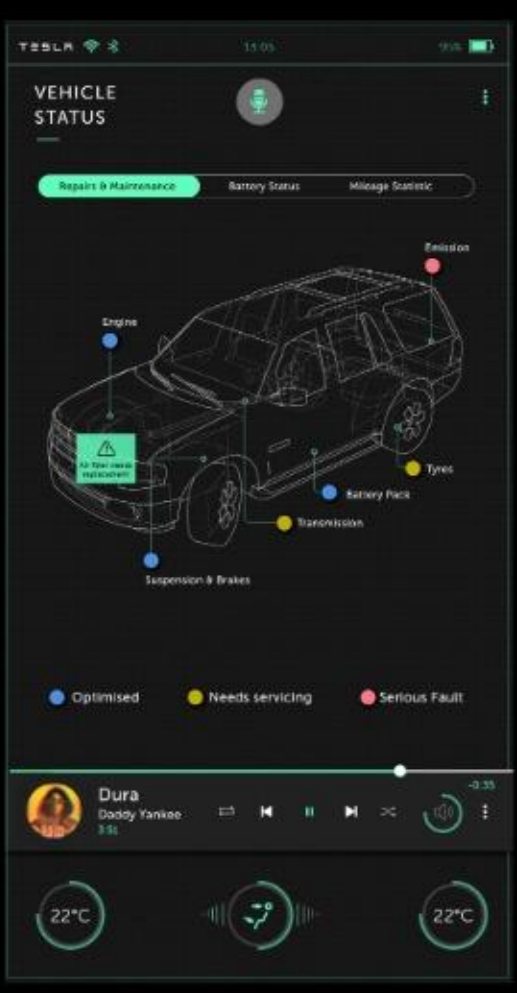
MAIN MENU



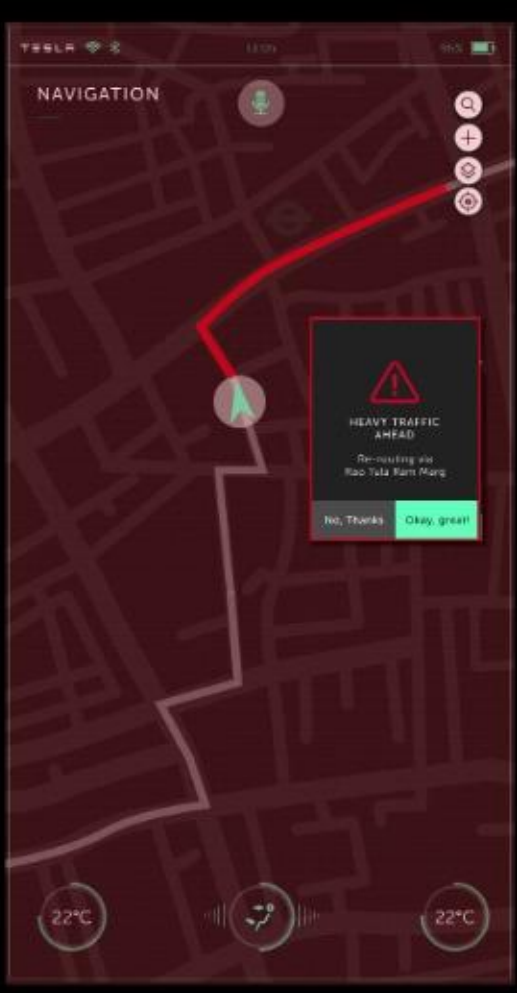
NAVIGATION



VEHICLE STATUS// REPAIRS & MAINTENANCE



ALERT WITHIN NAVIGATION



THANKYOU